

Dangers of Bayesian Model Averaging under Covariate Shift

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- 3 Proposed Solution: EmpCov Prior
- 4 Key Takeaways

The Ubiquitous Problem of Covariate Shift

Real-world machine learning challenge

Training and test distributions are rarely identical in practice:

$$p_{\text{train}}(x) \neq p_{\text{test}}(x)$$

but conditional distribution remains: $p(y|x)$ unchanged.

Applications affected:

- Medical diagnosis
- Autonomous driving
- Financial forecasting
- Natural language processing

Types of shift:

- Test data corruption
- Domain shift
- Spurious correlations
- Temporal drift

Key question: Which methods are robust to such shifts?

Bayesian Methods: Promise vs. Reality

Theoretical promise:

Bayesian Neural Networks (BNNs) provide **epistemic uncertainty**

Bayesian Model Averaging (BMA):

$$p(y|x) = \int p(y|x, w)p(w|\mathcal{D})dw$$

Should be robust to distribution shift

Captures multiple explanations for OOD data

Surprising reality:

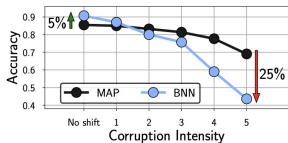
BNNs with high-fidelity HMC inference perform **poorly** under covariate shift

Often **underperform** simple MAP estimation

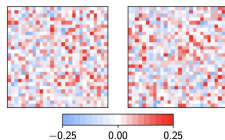
Particularly vulnerable to noise corruptions

This paper investigates **why** and proposes solutions

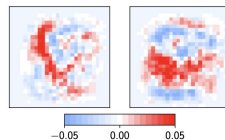
Bayesian neural networks under covariate shift



(a) ResNet-20, CIFAR-10-C



(b) BNN weights



(c) MAP weights

Bayesian Framework Review

Bayesian Neural Network

Prior: $p(w)$

Likelihood: $p(\mathcal{D}|w)$

Posterior via Bayes:

$$p(w|\mathcal{D}) = \frac{p(\mathcal{D}|w)p(w)}{\int p(\mathcal{D}|w')p(w')dw'}$$

Prediction via BMA:

$$p(y|x) = \int p(y|x, w)p(w|\mathcal{D})dw$$

MAP Estimation

$$w_{\text{MAP}} = \arg \max_w p(w|\mathcal{D})$$

Equivalent to:

$$\arg \max_w [\log p(\mathcal{D}|w) + \log p(w)]$$

where $\log p(w)$ acts as regularizer.

Theorem (Lemma 1 - Simplified)

Consider a BNN with first-layer weight w_{ij}^1 for input feature x^i . If $x_k^i = 0$ for all training examples $x_k \in \mathcal{D}$, and the prior factorizes:

$$p(W) = p(w_{ij}^1) \cdot p(W \setminus w_{ij}^1)$$

then:

$$p(w_{ij}^1 | \mathcal{D}) = p(w_{ij}^1)$$

The posterior equals the prior, and MAP sets w_{ij}^1 to the prior mode (typically 0).

Theorem (Proposition 2 - Key Points)

Suppose training inputs $x_1, \dots, x_n$ lie in an affine subspace:

$$\sum_{j=1}^m x_i^j c_j = c_0 \quad \forall i, \quad \text{with} \quad \sum_{j=0}^m c_j^2 = 1$$

For i.i.d. Gaussian prior $\mathcal{N}(0, \alpha^2)$:

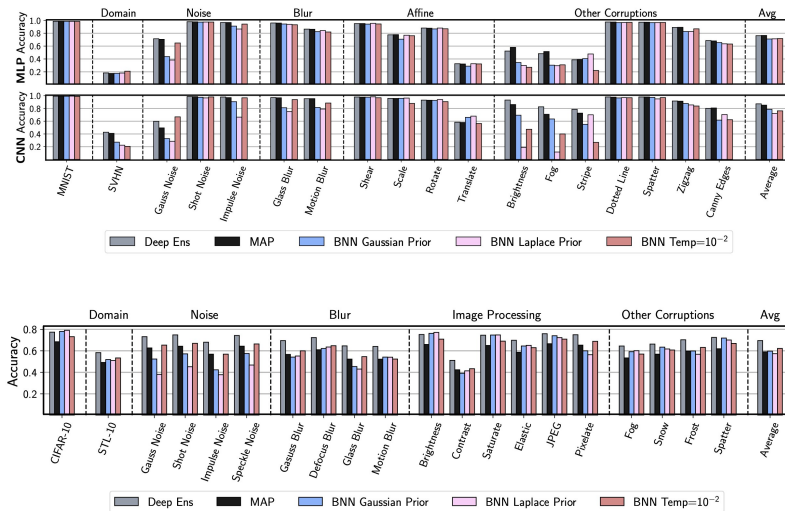
1° For neuron j in first layer, posterior of projection

$$w_j^c = \sum_{i=1}^m c_i w_{ij}^1 - c_0 b_j^1 \text{ coincides with prior}$$

2° MAP sets $w_j^c = 0$

3° For ReLU activations, BMA predictions are sensitive to test inputs outside the subspace

Another example of covariate shift



Proposed solution

The Empirical Covariance (EmpCov) prior for first-layer weights:

$$p(w^1) = \mathcal{N}(0, \alpha \Sigma + \epsilon I)$$

where:

$\Sigma = \frac{1}{n-1} \sum_{i=1}^n x_i x_i^\top$ is the empirical covariance of inputs

$\alpha > 0$ controls prior scale

$\epsilon > 0$ ensures positive definiteness

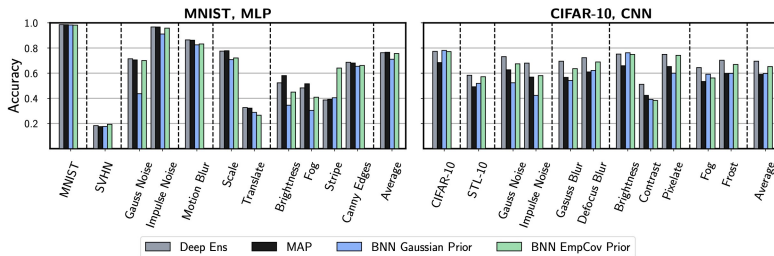
Eigenvectors of prior = Principal components of data

Prior variance along PC p_i : $\alpha \sigma_i^2 + \epsilon$

For zero-variance direction ($\sigma_i^2 = 0$): variance = ϵ (small)

This prevents BNN from sampling large random weights along low-variance directions

Method	Why More Robust to Covariate Shift
MAP (SGD)	Regularization (log prior) pushes irrelevant weights to zero
Deep Ensembles	Average of multiple MAP solutions, inheriting robustness
SWAG	Gaussian approximation centered around trajectory (near MAP)
MC Dropout	Implicit regularization through dropout during training
Variational Inference	Often collapses to MAP-like solutions due to Gaussian approximation
BNN (HMC)	Samples exact posterior, including random weights from prior for low-variance directions



Key Takeaways

- 1° **BNNs with exact inference (HMC) are vulnerable to covariate shift**, particularly noise corruptions
- 2° **Root cause**: Linear dependencies in training data cause posterior to equal prior in certain directions
- 3° **MAP and ensembles are more robust** due to regularization pushing irrelevant weights to zero
- 4° **EmpCov prior** significantly improves BNN robustness by aligning prior variance with data variance